

Michio Honda

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INTERESTS	Computer Networks, Operating Systems and Security	
EDUCATION	Keio University , Tokyo, Japan	
	Ph.D. in Graduate School of Media and Governance, Cyber Informatics Thesis: The Internet is not an Internet—Principles, Evasion and Implications for Transport Protocols	Apr 2009 – Mar 2012
	M.S. in Graduate School of Media and Governance, Cyber Informatics Thesis: Bidimensional-Probe Multipath Congestion Control for Shared Bottleneck Fairness	Apr 2007 – Mar 2009
	B.S. in Faculty of Environment and Information Studies Thesis: Fast Transport Layer Handover Using Single Wireless Interface	Apr 2003 – Mar 2007
EMPLOYMENT	University of Edinburgh, School of Informatics , Edinburgh, UK	
	Reader (Associate Professor)	Aug 2024 – present
	Lecturer in Networked Systems (Assistant Professor)	Jan 2020 – Jul 2024
	Senior researcher , NEC Laboratories Europe, Heidelberg, Germany	Nov 2016 – Dec 2019
	Software engineer , NetApp, Munich, Germany	Dec 2014 – Oct 2016
	Research scientist , NEC Laboratories Europe, Heidelberg, Germany	Jul 2012 – Nov 2014
OTHER EXPERIENCE	Visiting student researcher , University College London (UCL), London, UK Advisor: Prof. Mark Handley Focus: Middlebox, Multipath Transport Protocol.	
	Research intern , Nokia Research Center, Espoo, Finland Advisor: Dr. Lars Eggert Focus: Multipath Transport Protocol.	Apr 2010 – Sep 2010 Jul 2008 – Jan 2009
AWARDS	Google Research Scholar Award *	Apr 2022
	Facebook Research Award *	Aug 2021
	Best paper award , ACM SOSR'15 *	Jun 2015
	Community award , USENIX NSDI'12	Apr 2012
	IETF/IRTF Applied Networking Research Prize (ANRP)	Nov 2011* and Jan 2026
	* as an individual recipient or lead author	
SELECTED PROJECTS	Secure datacenter transport protocol SMT is a new encrypted transport protocol designed for low-latency datacenter transports. It refines TLS with per-message record sequence number spaces to avoid head-of-line blocking, while preventing replay attacks. Looma enables low-latency TLS handshake with post-quantum resistance and mutual authentication in the cloud, using an online/offline signature scheme.	
	Host stack support for traffic analysis attack defenses Traffic analysis attacks like website fingerprinting is a growing threat to the encrypted Internet traffic. Stob is a framework that integrates traffic obfuscation directly in the host network stack. Its design rationale is that packet-level features that the attacks exploit, like sizes, bursts and timing, are determined at the transport layer or below.	Oakland'26, NDSS'26 HotNets'25
	Flexible TCP load balancing Prism is a new layer 7 load balancing (L7LB) architecture that enables the backend server to return the response to the client directly, bypassing the L7LB device. Prism uses TCP connection migration with the aid of programmable switches; XO enables similar features to Prism but without switch support, and integration with real applications like Nginx and Ceph.	NSDI'26, NSDI'21
	Network stack design for modern hardware PASTE is a network stack that offers unified abstractions of network and non-volatile main memory. It fills the gap between the storage and network stacks designed in isolation, and solves the problem with the costs of moving and transforming data between these stacks that are significant for non-volatile main memory that offers fast, byte-addressable persistence.	ATC'25, HotNets'21, NSDI'18, HotNets'16, ATC'16
	Scalable virtual networking mSwitch solves the scalability problem of existing software switches that is crucial to consolidate a large number of VMs or virtualized network functions by a novel packet forwarding algorithm and streamlined data path. It was initially designed for ClickOS, a tiny unikernel that runs Click, and MultiStack, a framework that runs multiple user-space network stacks.	SOSR'15, CCR'14, NSDI'14, SoCC'17
	Middlebox measurement for TCP extensibility	IMC'11, NSDI'12

This work was motivated by exploring viable design of Multipath TCP. It transmits various non-existent TCP traffic that mimics possible future TCP extensions to our server and examines on-path actions to the packets. This is the first work that examines in-depth middlebox behaviour prevalent in the Internet.

PROFESSIONAL SERVICE	UNIVERSITY OF EDINBURGH	
	People and Culture committee	2022–present
	Programming Club, organizer	2020–2024
	PROFESSIONAL SOCIETY	
	ACM SIGOPS, CARES Committee co-chair	2023–present
	IRTF ANRP Award Committee	2022–present
	ACM SIGOPS, CARES Committee	2021–2022
	CONFERENCE PROGRAM COMMITTEE	
	USENIX OSDI (2025–2027), NSDI (2023, 2026), ACM EuroSys (2021, 2024–2026), ACM SIGCOMM (2024–2026), ACM APSYS (2024–2025), ACM HotNets (2024), ACM ASPLOS (2024), USENIX/ACM ATC (2017, 2018, 2020–2024, 2026), ACM HotStorage (2022–2025), ACM/IEEE ANCS (2018, 2021 (co-chair)), ACM CoNEXT (2021–2022), ACM/IEEE SC (2019), ACM SOSR (2018), ACM EuroDW (2018, ACM/IEEE ToN (2017–2018), ACM SOSP poster (2013)	
TEACHING	Introduction to Programming (Informatics) Summer , University of Edinburgh	Jul 2021 and 2022
	Computer Communications and Networks , University of Edinburgh	Fall 2021–present
	Data Structures and Programming , Keio University	Fall 2011
	Fundamentals of Information Technology , Keio University	Spring 2011
MENTORING	UNIVERSITY OF EDINBURGH	
	Shuo Li, PhD supervision (graduated)	Nov 2021–Jul 2026
	Xinshu Ma, PhD supervision (degree awarded)	Jul 2023–present
	Tianyi Gao, PhD supervision	Dec 2022–present
	Elisaveta Lavrentieva, PhD supervision (equal co-supervision with Marc Juarez)	Sep 2024–present
	Michael Zhang, Internship mentoring (ICSA Summer Internship)	Summer 2024
	Eugenio Luo, Internship mentoring (EPSRC Vacation Internship)	Summer 2024
	Tianyi Gao, MSc and intern supervision	Spring 2022
	Steven W. D. Chien, Postdoc → assistant professor at the University of St Andrews.	Mar 2022–Sep 2025
	Shinichi Awamoto, PhD supervision	Jan 2021–Jan 2024
	NETAPP	
	Nanako Momiyama, BSc thesis supervision, Keio University	Fall 2016
	Yutaro Hayakawa, BSc thesis intern at NetApp, Keio University	Fall 2016
	Kenichi Yasukata, MSc thesis intern at NetApp, Keio University	Fall 2015
	NEC LABS EUROPE	
	Shinichi Awamoto, intern at NEC, Tokyo University	Fall 2019
	Yutaro Hayakawa, MSc thesis intern at NEC, Keio University	Fall 2018
	Nanako Momiyama, intern at NEC	Spring 2017
	Kenichi Yasukata, mentor at NEC	Fall 2016
GRANTS	GLOBAL	
	NetApp Faculty Fellowship , \$50K, sole PI	Jan 2023
	Towards Generic, Encrypted Datacenter Transport	
	Google Research Scholar Award , \$60K, sole PI	May 2022
	Upcycling Packets as Persistent In-Memory Data Structures	
	Facebook Research Award , \$50K, sole PI	Nov 2021
	Flexible transport scale-out with modern NICs	
	REGIONAL (UK/EU)	
	EPSRC Core Equipment Award , £50K (my share), Co-I (my part) Systems Research Testbed	Jan 2025 – Jun 2026
	Royal Society Research Grant , £20K, sole PI	Oct 2024 – Oct 2025
	Confidential Computing at a Scale	
	EPSRC Core Equipment Award , £35K (my share), Co-I (my part) Systems Research Testbed	Jan 2023 – Mar 2023
	EPSRC New Investigator Award , £385K, sole PI	Apr 2022 – Mar 2025
	NetPM: Co-designing Data Management and Networking Principles for Persistent Memory	
	NCSC RISE Proof-of-Concept , £45K, co-PI	Oct 2021 – Mar 2022
	Follow on Project: Gupt - A Hardware Assisted Secure and Private Data Analytics	

OTHER HONORS	Nominee for Teaching Award , Outstanding Course category, University of Edinburgh	Apr 2023
STUDENT SCHOLARSHIP	Research Fellowship for Young Scientists (DC1) Japan Society for the Promotion of Science, 9.2M JPY	Apr 2009 – Mar 2012
	Excellent Young Researcher Overseas Visit Program Japan Society for the Promotion of Science, 1M JPY	Apr 2010
	Young Leader Scholarship Keio University, 1M JPY	Apr 2009
SELECTED PUBLICATIONS	* denotes equal contribution or supervision role.	
	Tianyi Gao, Xinshu Ma, Suhas Narreddy, Eugenio Luo, Steven W. D. Chien and Michio Honda , “ <i>Designing Transport-Level Encryption for Datacenter Networks</i> ”, IEEE Symposium on Security and Privacy (S&P “Oakland”), May 2026. IETF/IRTF Applied Networking Research Prize	
	Shuo Li*, Steven W. D. Chien*, Tianyi Gao and Michio Honda , “ <i>Remote TCP Connection Offload and Applications</i> ”, USENIX Symposium on Networked Systems Design and Implementation (NSDI), May 2026.	
	Xinshu Ma and Michio Honda , “ <i>Looma: A Low-Latency PQTLS Authentication Architecture for Cloud Applications</i> ”, ISOC Network and Distributed System Security Symposium (NDSS), Feb 2026.	
	Elisaveta Lavrentieva, Marc Juarez* and Michio Honda* , “ <i>Rethinking the Role of Network Stacks for Website Fingerprinting Defenses</i> ”, ACM Workshop on Hot Topics in Networks (HotNets), Nov 2025.	
	Steven W. D. Chien, Kento Sato, Artur Podobas, Niclas Jansson, Stefano Markidis and Michio Honda , “ <i>ParaLog: Consistent Host-Side Logging for Parallel Checkpoints</i> ”, ACM Symposium on Cloud Computing (SoCC), Nov 2025.	
	Shinichi Awamoto and Michio Honda , “ <i>Opening Up Kernel-Bypass TCP Stacks</i> ”, USENIX Annual Technical Conference (ATC), Jul 2025.	
	Michio Honda , “ <i>Packets as Persistent In-Memory Data Structures</i> ”, ACM Workshop on Hot Topics in Networks (HotNets), Nov 2021.	
	Yutaro Hayakawa, Michio Honda , Douglas Santry and Lars Eggert, “ <i>Prism: Proxies without the Pain</i> ”, USENIX Symposium on Networked Systems Design and Implementation (NSDI), Apr 2021.	
	Shinichi Awamoto, Erich Focht and Michio Honda , “ <i>Designing a Storage Software Stack for Accelerators</i> ”, USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage), Jul 2020.	
	Maurice Bailieu, Jörg Thalheim, Pramod Bhatotia, Christof Fetzer, Michio Honda and Kapil Vaswani, “ <i>Speicher: Securing LSM-based Key-Value Stores using Shielded Execution</i> ”, USENIX Conference on File and Storage Technologies (FAST), Feb 2019.	
	Salvatore Pontarelli, Roberto Bifulco, Marco Bonola, Carmelo Cascone, Marco Spaziani, Valerio Bruschi, Davide Sanvito, Giuseppe Siracusano, Antonio Capone, Michio Honda , Felipe Huici and Giuseppe Bianchi, “ <i>FlowBlaze: Stateful Packet Processing in Hardware</i> ”, USENIX Symposium on Networked Systems Design and Implementation (NSDI), Feb 2019.	
	Michio Honda , Giuseppe Lettieri, Lars Eggert and Douglas Santry, “ <i>PASTE: A Network Programming Interface for Non-Volatile Main Memory</i> ”, USENIX Symposium on Networked Systems Design and Implementation (NSDI), Apr 2018.	
	Kenichi Yasukata, Felipe Huici, Vincenzo Maffione, Giuseppe Lettieri and Michio Honda , “ <i>HyperNF: Building a High Performance, High Utilization and Fair NFV Platform</i> ”, ACM Symposium on Cloud Computing (SoCC), Sep 2017.	
	Simon Kuenzer, Anton Ivanov, Filipe Manco, Jose Mendes, Yuri Volchkov, Florian Schmidt, Kenichi Yasukata, Michio Honda and Felipe Huici, “ <i>Unikernels Everywhere: The Case for Elastic CDNs</i> ”, ACM International Conference on Virtual Execution Environments (VEE), Apr 2017.	
	Michio Honda , Lars Eggert and Douglas Santry, “ <i>PASTE: Network Stacks Must Integrate with NVMM Abstractions</i> ”, ACM Workshop on Hot Topics in Networks (HotNets), Nov 2016.	
	Kenichi Yasukata, Michio Honda , Douglas Santry and Lars Eggert, “ <i>StackMap: Low-Latency Networking with the OS Stack and Dedicated NICs</i> ”, USENIX Annual Technical Conference (ATC), Jun 2016.	
	Michio Honda , Felipe Huici, Giuseppe Lettieri and Luigi Rizzo, “ <i>mSwitch: A Highly-Scalable, Modular Software Switch</i> ”, ACM SIGCOMM Symposium on SDN Research (SOSR), Jun 2015. Best paper award	
	Michio Honda , Felipe Huici, Costin Raiciu, Joao Araujo and Luigi Rizzo, “ <i>Rekindling Network Protocol Innovation with User-Level Stacks</i> ”, ACM SIGCOMM Computer Communication Review (CCR), Apr 2014.	

Joao Martins, Mohamed Ahmed, Costin Raiciu, Vladimir Olteanu, **Michio Honda**, Roberto Bifulco and Felipe Huici, “ClickOS and the Art of Network Function Virtualization”, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Apr 2014.

Costin Raiciu, Christoph Paasch, Sebastien Barre, Alan Ford, **Michio Honda**, Fabien Duchene, Olivier Bonaventure and Mark Handley, “How Hard Can It Be? Designing and Implementing a Deployable Multipath TCP”, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Apr 2012. **Community Award**

Michio Honda, Yoshifumi Nishida, Costin Raiciu, Adam Greenhalgh, Mark Handley and Hideyuki Tokuda, “Is it Still Possible to Extend TCP?” ACM Internet Measurement Conference (**IMC**), Nov 2011. **IETF/IRTF Applied Networking Research Prize**

Michio Honda, Yoshifumi Nishida, Pasi Sarolahti and Lars Eggert, “Multipath Congestion Control for Shared Bottleneck” International Workshop on Protocols for Future, Large-Scale & Diverse Network Transports (**PFLDNeT**), May 2008.

Michio Honda, Jin Nakazawa, Yoshifumi Nishida, Masahiro Kozuka and Hideyuki Tokuda, “A Connectivity-Driven Retransmission Scheme Based On Transport Layer Readdressing”, IEEE International Conference on Distributed Computing Systems (**ICDCS**), Jun 2008.

INTERNET DRAFTS	Xinshu Ma, Michio Honda and Colin Perkins, “Looma: Low-Latency Post-Quantum Authentication for TLS 1.3 in Datacenters”, draft-ma-cfrg-looma-00, Mar 2026.		
OPEN SOURCE CONTRIBUTION (by myself, not mentees)	PASTE	https://micchie.github.io/paste/	
	netmap	mSwitch and various features https://github.com/luigirizzo/netmap	
	MultiStack	https://github.com/sysml/multistack	
	Linux kernel	SCTP extensions https://www.kernel.org/	
	FreeBSD kernel	mSwitch and SCTP extensions https://www.freebsd.org/	

[CV compiled on 2026-08-26]